Solution Proposal

Subtitle for Document

Prepared for: Customer Name

April 23, 2024
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Document Information

Document Title	**Solution Proposal - Subtitle for Document**
Document Owner	[redacted]
Document Classification	Classified

Document Change Record

Date	Author	Title	Version	Change Reference
04/23/2024	[redacted]		1.0	Initial version

Document Reviewers

Date	Reviewer	Title	Version

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Overview

Who We Are

HYVE is a highly specialized Data Analytics and AI technology service provider that offers cutting-edge solutions to help customers maximize the value of their data and harness the power of AI technologies to drive their business forward. Our core focus is on delivering bespoke data-driven insights and actionable recommendations that enable our clients to make informed decisions and achieve their business objectives.

HYVE has consistently demonstrated its expertise in delivering cutting-edge analytics solutions across a diverse range of projects and industries. Our extensive experience encompasses the development of cloud-based performance tracking systems, interactive dashboards, planning and monitoring tools, data warehouse implementation, and advanced analytics solutions tailored to clients' unique needs.

We pride ourselves on our ability to seamlessly integrate with various data sources and systems, optimize business intelligence, and streamline decision-making processes for our clients.

Our diverse portfolio of successful projects highlights our commitment to partnering with organizations across various sectors, including government entities, public sector organizations, and private enterprises, to harness the power of data-driven insights and drive tangible results.

What We Do

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Executive Summary

KFH seeks to deliver a step-change in customer engagement and operational excellence by deploying an AI-powered Robo Assistant and a unified Customer 360 on a resilient, scalable Red Hat OpenShift foundation. HYVE proposes an end-to-end, enterprise-grade solution that unifies mobile, web, ATMs, kiosks, XTMs, and branch interactions into a single, consistent journey; powers real-time personalization and next-best-actions with explainable AI; orchestrates omnichannel campaigns; and integrates securely with core banking, CRM, and the data lake through real-time streaming. Our approach is designed to reduce customer effort, increase digital adoption and cross-sell, preempt churn, and improve service productivity—while meeting stringent security, privacy, and regulatory requirements.

1. Vision and Value Proposition

The proposed solution fuses a 360-degree customer intelligence layer with an omnichannel conversational experience, enabling KFH to anticipate needs, personalize at scale, and streamline service across every touchpoint. By containerizing microservices on Red Hat OpenShift and employing a robust service mesh, autoscaling, and CI/CD, we ensure agility, high availability, and efficient lifecycle management for AI workloads. Real-time streaming (for example, using Apache Kafka) captures behavioral signals—app events, transactions, service interactions—to feed a governed feature store powering predictive models. This cohesive architecture lifts conversion and retention, enhances customer satisfaction, and accelerates time-to-market for new products and journeys.

Our value proposition is grounded in measurable outcomes. We target meaningful improvements across the funnel: faster onboarding and product fulfillment, uplift in cross/upsell driven by context-aware offers, and reduced churn risk through proactive re-engagement. Operationally, we reduce average handling time in assisted channels, deflect repetitive queries to the Robo Assistant, and improve campaign ROI through precise targeting and experimentation. At every step, explainable AI (XAI) and bias governance ensure transparency, fairness, and auditability—building trust with customers, business stakeholders, and regulators.

2. Unified Omnichannel Experience and Customer 360

A single customer identity and profile underpins every channel experience. We build golden profiles by resolving identities across mobile banking, web, CRM, core banking, and device endpoints (ATMs, kiosks, XTMs), unifying demographic, transactional, behavioral, and interaction data. This Customer 360 becomes the authoritative source for personalization and analytics, codified into a real-time feature store that exposes curated features to models and decision services. The result is a continuous journey that “remembers” customer context—where they started, what they attempted, and what they need next—irrespective of channel.

In practice, a customer browsing a home financing product on the mobile app can continue in a branch or through the Robo Assistant without repeating steps. The assistant understands prior events, retrieves the relevant knowledge securely, and tailors guidance. When a high-value customer triggers a churn-risk pattern (for example, decreased balances and declining engagement), the system can surface a retention action across the next touchpoint—push notification, in-app message, or a proactive relationship manager callback—backed by transparent rationale and approvals.

3. AI-Powered Robo Assistant with Real-Time Personalization

The Robo Assistant provides secure, multilingual (Arabic/English) conversational support, spanning FAQs, transactional assistance, product discovery, and guided service journeys. It integrates with core systems through hardened APIs and leverages retrieval-augmented generation (RAG) to ground responses in approved, versioned knowledge articles, forms, and policy documents. Context continuity allows the assistant to reflect prior interactions and user preferences, while guardrails enforce compliance with KFH policy and regulatory norms.

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Real-time personalization enriches the conversation and surrounding digital UI elements. Next-best-action and next-best-offer models propose contextual nudges—such as completing a partially started application, opting into a savings plan, or educating customers on debit card security features. For assisted channels, the same intelligence powers agent assist: branch staff and contact center agents receive recommended actions, scripted responses, and cross-sell prompts with supporting explanations, ensuring consistency between human and virtual interactions. When needed, customers can be seamlessly handed off to a human agent with full context transfer, reducing friction and increasing first-contact resolution.

4. Predictive Insights, Churn Prevention, and Re-Engagement

Predictive churn models continuously assess risk using multivariate signals: login frequency, feature usage depth, inbound service patterns, balance volatility, and complaint history. Propensity models estimate the likelihood of response to specific products or actions—installment plans, term deposits, credit cards—factoring eligibility, risk segments, and past behavior. These feed into a governed decision engine that sets thresholds, prioritizes actions, and enforces fairness constraints and business rules. The output is a targeted retention or growth strategy, delivered across digital and assisted touchpoints.
Re-engagement plays are orchestrated for inactive users: frictionless sign-in prompts, reminders to complete KYC or deferred applications, fee waiver offers within policy limits, and education flows highlighting feature value (for example, budgeting tools or card controls). Through A/B/n testing and uplift modeling, the system learns which treatments work for each persona and context, improving conversion while minimizing cost of contact. Every model prediction is logged with its features and rationale for auditability, and can be explained to customers and internal stakeholders where appropriate.

5. Omnichannel Campaign Orchestration and Journey Management

A centralized journey builder coordinates campaigns across push, SMS, email, in-app, web banners, ATM screen messaging, XTM prompts, and branch/agent scripts. Journeys are triggered by events (for example, first salary credit, dormant account awakening, or declined application), enriched by Customer 360 attributes and real-time model scores. Frequency caps, consent management, and channel preferences are enforced at orchestration time to protect customer experience and comply with privacy mandates.
Marketers can iterate quickly through controlled experiments, using holdout groups, guardrails, and cross-channel attribution to measure incremental lift. Success metrics—activation rates, product uptake, revenue contribution, and cost per acquisition or retention—are visible in role-based dashboards. Over time, journey performance feeds back into the feature store and models, creating a virtuous cycle of continuous improvement grounded in real-world outcomes.

6. Red Hat OpenShift Foundation: Scalability, Reliability, and Speed

The platform is built on containerized microservices orchestrated by Red Hat OpenShift, enabling declarative deployments, GitOps-driven CI/CD, and consistent policy enforcement. A service mesh provides secure, observable service-to-service communication, while autoscaling accommodates peak demand—such as mass salary days or seasonal campaigns—without service degradation. The design supports high availability through multi-zone clustering and offers both active-active and active-passive disaster recovery options, with clear RTO/RPO targets.
Observability stacks deliver centralized logging, metrics, and distributed tracing to pinpoint bottlenecks and optimize cost-performance. Secrets management and configuration-as-code ensure that environment drift is minimized and that promotions from development to production are controlled, auditable, and repeatable. The net effect is faster release cycles, lower operational risk, and reduced total cost of ownership compared with monolithic or manually managed environments.

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7. Real-Time Integration, Data Streaming, and Governance

The solution integrates via an API gateway and secure adapters to core banking, CRM, digital channels, ATMs, kiosks, XTMs, and the enterprise data lake. Real-time streaming pipelines capture events and state changes, keeping Customer 360 and the feature store fresh within seconds. Batch ingestion supports historical data loads and periodic enrichment. Data contracts and schema registries maintain compatibility, while idempotent patterns and retry logic preserve consistency under transient failures. Data governance is integral. We enforce lineage tracking, data quality checks, and access controls aligned to roles and attributes (RBAC/ABAC). Personally identifiable information is masked or tokenized where appropriate, and every dataset includes clear ownership, retention policies, and approved use cases. Model governance extends these controls to the AI layer, with model registry, approvals, performance monitoring, drift detection, and rollback processes, ensuring that only validated models influence customer-facing decisions.

8. Security, Privacy, Compliance, and Explainable AI

Security and privacy are engineered by design. Encryption is applied in transit and at rest; strong identity and access management guards administrative and service access; and comprehensive audit logging satisfies internal and external oversight. The platform aligns with KFH policies and applicable regulatory requirements, including privacy and data localization mandates. Data minimization, consent enforcement, and purpose limitation are enforced in orchestration and analytics workflows. Explainable AI is standard for all models influencing customer outcomes. We employ interpretable techniques and post-hoc explainers (for example, SHAP-like methods) to expose the drivers behind predictions, with thresholds for human review on sensitive decisions. Bias testing evaluates protected attributes and proxy effects, triggering mitigation through reweighting, constrained optimization, or rule overlays. A human-in-the-loop process governs high-impact actions, ensuring accountability and ethical use. These safeguards protect customers and KFH while enabling confident, compliant innovation.

Persona	**Key Problem Statements**	**Targeted Business Outcomes**
Retail Customer	Fragmented experience across mobile/web/ATMs; repetitive authentication and re-entry of information; slow resolution of routine queries	Seamless context continuity; faster self-service with Robo Assistant; reduced time-to-completion for top tasks; higher satisfaction and adoption
Premier/Wealth	Generic offers not aligned to portfolio goals; limited proactive insights; high sensitivity to service quality	Personalized advice prompts; tailored offers and alerts; relationship manager assist with AI insights; improved retention and NPS
SME	Complex onboarding and product discovery; limited guidance on working capital solutions	Guided journeys for onboarding/loans; targeted propositions based on cashflow patterns; faster approvals and servicing
Branch Staff	Disconnected systems and manual lookups; inconsistent cross-sell prompts	Agent assist with next-best-actions; unified customer snapshot; reduced average handling time and higher conversion

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Contact Center	High volume of repetitive queries; limited visibility into digital context	Deflection to Robo Assistant; full interaction history; improved first-contact resolution and reduced queue times
Marketing	Batch-only campaigns with low relevance; limited attribution	Real-time, event-triggered journeys; A/B/n testing and holdouts; measurable lift and improved ROI
IT/Ops	Monolithic releases; scalability constraints; limited observability	OpenShift microservices with CI/CD; autoscaling and service mesh; full-stack observability and faster recovery
Risk/Compliance	Opaque AI decisions; model drift and bias risk; fragmented audit trails	XAI with bias governance; model registry and approvals; comprehensive logging, lineage, and auditability

9. Key Capabilities and Measurable Benefits

The solution's components are engineered to deliver tangible, trackable improvements across customer experience, revenue, and efficiency. The Robo Assistant reduces assisted-channel volume by resolving high-frequency intents instantly, while escalation paths preserve service quality for complex cases. Customer 360 fuels individualized experiences that increase relevance and trust, translating into higher product uptake and digital engagement. Predictive models address both sides of the growth equation—preventing avoidable churn and catalyzing next-best actions—while orchestration ensures every insight is operationalized promptly and consistently.

Capability	**Description**	**Measurable Benefit**
Customer 360 & Feature Store	Golden profiles, real-time features from streaming and batch sources, governed access	Higher personalization precision; reduced time-to-model by streamlining data prep
AI Robo Assistant	Secure, multilingual conversational support with RAG and channel continuity	Lower average handling time; increased self-service rate; consistent service quality
Predictive Analytics (Churn, Propensity, NBA)	Models driving retention and tailored offers with XAI	Reduced churn rate; increased cross/upsell conversion; transparent decisioning
Omnichannel Orchestration	Event-triggered journeys, experimentation, and attribution	Higher marketing ROI; faster iteration; improved activation and re-engagement
OpenShift Microservices	Autoscaling, service mesh, CI/CD, HA/DR	Improved uptime; faster releases; cost-efficient scaling for AI workloads
Security & Compliance	Encryption, IAM, auditing, privacy controls	Reduced risk of incidents; regulator-ready audit trails; policy alignment

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10. Delivery Approach and Milestones at a Glance

HYVE will deliver through iterative, transparent sprints with stage gates. Early phases establish governance, personas, and outcomes; mid-phases implement the platform, Robo Assistant, 360 data foundations, streaming pipelines, and models; and final phases focus on hardening, UAT, enablement, and go-live with hypercare. This cadence ensures demonstrable value at each increment and de-risks downstream integration and adoption. While a detailed plan follows in subsequent sections, an indicative summary of phases and deliverables is provided for executive alignment.

Phase	Duration	Key Deliverables
Phase 1: Initiation & Governance	2 weeks	Stakeholder alignment; governance forums; success metrics baseline; secured environments
Phase 2: Discovery & Personas	3–4 weeks	Persona journeys; problem-to-outcome maps; prioritized use cases; data source inventory
Phase 3: Architecture & Integration Design	4–6 weeks	Target architecture; API/streaming designs; security baseline; runbooks outline
Phase 4: Build Foundations	8–10 weeks	OpenShift services; Customer 360 and feature store; streaming pipelines; initial Robo Assistant
Phase 5: Models & Orchestration	4–6 weeks	Churn/propensity/NBA models; XAI/bias testing; campaign journeys; MLOps pipelines
Phase 6: E2E Integration & Hardening	3–4 weeks	Performance tests; HA/DR validation; security hardening; monitoring and alerts
Phase 7: UAT, Training & Go-Live	2–3 weeks	UAT sign-off; enablement; production deployment; hypercare and handover

11. Why HYVE for KFH

HYVE combines domain-centric data and AI expertise with disciplined engineering on enterprise platforms. Our approach emphasizes sprint-to-value delivery, rigorous governance, and measurable business outcomes. We design for the realities of financial services—security, privacy, resilience, and auditability—without compromising the speed and agility needed to compete in digital channels. By partnering with KFH, we will co-create a platform and operating model that delivers immediate wins and forms a durable foundation for future innovation, from advanced credit decisioning to proactive financial wellness and beyond.

In summary, HYVE's AI-powered Robo Assistant and Customer 360 on Red Hat OpenShift provides KFH with a secure, scalable, and explainable platform to personalize experiences in real time, orchestrate effective journeys across every channel, and translate data into sustained growth and service excellence. Through proven architecture, agile delivery, and rigorous governance, we will help KFH achieve rapid impact while safeguarding trust and compliance.

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Engagement Approach

HYVE's engagement approach is structured to deliver early value while de-risking complexity through iterative, outcome-driven delivery. The program is organized into seven phases from initiation to hypercare, each with clear entry/exit criteria, stage-gate reviews, and measurable outcomes. The cadence is based on two-week sprints with continuous stakeholder alignment, rigorous governance, and transparent reporting. Our approach is tailored to KFH's objectives for an AI-powered Robo Assistant, a unified Customer 360 with a governed feature store, real-time personalization and campaign orchestration, OpenShift-based microservices, real-time streaming integrations, and enterprise-grade security, compliance, and explainable AI.

1. Governance, Kickoff, and Planning (Phase 1: 2 weeks)

The engagement begins with the creation of a joint governance framework that ensures clarity of scope, accountability, and decision velocity. HYVE and KFH establish a program steering committee, a delivery working group, and a model governance board. The kickoff aligns business and technical stakeholders on objectives, key results, and acceptance criteria. Initial activities include environment readiness for development and testing, access provisioning, risk baseline, and a communication plan. We define a shared definition of done for work items, documentation standards, and quality gates for code, models, data, and security. To accelerate execution, we set up a secure collaboration environment, repositories, and a backlog structure that maps problem statements to outcomes, epics, and user stories. The result of Phase 1 is a mobilized program with an agreed roadmap, RACI, and an operational governance rhythm with stage-gate checkpoints.

Milestone Name	**Target Date**	**Success Criteria**	**Dependencies**
Program Kickoff Complete	Week 2	Steering committee and working group established; communication plan approved	Stakeholder availability
Governance & RACI Baseline	Week 2	RACI signed off; decision and escalation pathways documented	Kickoff completion
Environments & Access Ready	Week 2	Dev/test OpenShift cluster namespaces, repos, and CI/CD pipelines initialized	IT/Ops provisioning
Role	**Name**	**Years of Experience**	**Key Responsibilities**
Program Director (HYVE)	To be nominated	18	Executive oversight, stakeholder alignment, risk governance, stage-gate approvals
Engagement Manager (HYVE)	To be nominated	12	Delivery orchestration, sprint planning, reporting, RAID management
Enterprise Architect (HYVE)	To be nominated	15	Target architecture, integration patterns, non-functional design, HA/DR

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Data & AI Lead (HYVE)	To be nominated	14	Feature store design, model strategy, XAI governance, MLOps
OpenShift Platform Lead (HYVE)	To be nominated	11	Cluster setup, GitOps, service mesh, observability
Security & Compliance Lead (HYVE)	To be nominated	13	Security baseline, IAM, audit, privacy controls
Product Owner (KFH)	Assigned by KFH	10	Value prioritization, backlog ownership, acceptance criteria
Channel/CRM Owner (KFH)	Assigned by KFH	9	Channel integration scope, customer journey alignment
Risk & Compliance Rep (KFH)	Assigned by KFH	12	Policy alignment, approvals, audit readiness

2. Persona-Led Discovery and Outcomes Mapping (Phase 2: 3–4 weeks)

Phase 2 focuses on translating KFH's problem statements into measurable outcomes through persona-driven discovery. We conduct structured interviews, journey walkthroughs, and in-situ observations across Retail, Premier/Wealth, SME, Branch Staff, Contact Center, Marketing, IT/Ops, and Risk/Compliance. For each persona, we define top tasks, pain points, operational constraints, and regulatory considerations. We produce current and target state journey maps that emphasize context continuity across mobile, web, ATMs, kiosks, XTMs, and branches. In parallel, we perform a data source inventory and assess integration readiness for core banking, CRM, channel applications, and the data lake. Non-functional requirements are documented, including performance SLOs, availability SLAs, RTO/RPO, data residency, and privacy obligations. The output is a prioritized backlog with quantifiable KPIs, such as deflection rate targets for the Robo Assistant, churn rate reduction windows, and conversion uplift goals for next-best-action offers.

Deliverable	**Description**	**Format**	**Delivery Date**
Persona Journey Maps	Target journeys with pain points, triggers, and KPIs	PDF + BPMN	Week 5–6
Problem-to-Outcome Matrix	Mapping of issues to measurable outcomes and acceptance thresholds	Spreadsheet	Week 6
Source System Inventory	APIs, schemas, event catalogs, SLAs, and owners	Spreadsheet + Confluence	Week 6
NFR & Compliance Baseline	Performance, HA/DR, privacy, data localization requirements	Document	Week 6

3. Target Architecture and Integration Design (Phase 3: 4–6 weeks)

In Phase 3, we design the end-to-end architecture centered on Red Hat OpenShift for container orchestration, GitOps CI/CD, and service mesh. The integration layer encompasses an API gateway with

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adapters into mobile and web channels, CRM, core banking, ATMs, kiosks, XTMs, and the enterprise data lake. We define event-driven patterns using Kafka-based streaming for real-time signals, complemented by batch ingestion for historical data. The Customer 360 lakehouse and governed feature store provide golden profiles and curated features for models and decision services. Security architecture includes RBAC/ABAC, network segmentation, secrets management, encryption in transit and at rest, and centralized audit logging. We also design model governance with a registry, lineage, XAI, and bias monitoring. Reliability is engineered via multi-zone clustering, active-active or active-passive DR, and clear RTO/RPO.

Component	Technology	Version	Purpose
Container Orchestration	Red Hat OpenShift	Latest LTS	Deploy and manage microservices with policy enforcement
Service Mesh	OpenShift Service Mesh (Istio-based)	Current	mTLS, traffic management, observability
CI/CD & GitOps	Tekton + Argo CD	Current	Declarative pipelines and environment promotions
API Gateway	3scale or equivalent	Current	API management, throttling, security, analytics
Event Streaming	Apache Kafka (managed or on OpenShift)	Current	Real-time events and integration backbone
Customer 360 Lakehouse	Cloud-native or on-prem lakehouse	Current	Unified profiles and analytics workspace
Feature Store	Open-source or enterprise feature store	Current	Governed, reusable model features
MLOps	MLflow + model registry + feature pipelines	Current	Model lifecycle, lineage, deployment
Observability	Prometheus, Grafana, EFK/ELK, Jaeger	Current	Metrics, logs, tracing, alerting
Security & IAM	Vault/Secrets, SSO, RBAC/ABAC	Current	Secrets, identity, access control

4. Iterative Build Sprints: Foundations and MVPs (Phase 4: 8–10 weeks)

Phase 4 delivers the core platform and MVP functionality through multiple two-week sprints. We implement OpenShift namespaces, cluster policies, ingress/egress controls, and service mesh. CI/CD pipelines enable automated builds, tests, scans, and progressive delivery across environments. Integration adapters connect to channels, CRM, core banking, and devices, while streaming pipelines start capturing real-time events. The Customer 360 data model is finalized and ingested with initial golden profiles; the feature store exposes curated, versioned features for models and decisioning. The Robo Assistant MVP focuses on high-frequency intents in Arabic and English, backed by retrieval-augmented generation with approved knowledge sources, secure session handling, and hand-off to human agents with full context continuity. Omnichannel campaign orchestration is introduced with event-triggered journeys, consent enforcement, and basic A/B testing. By the end of this phase, the

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program demonstrates working software that addresses priority personas and shows measurable value in test environments.

Resource	Allocation %	Duration	Cost
OpenShift Platform Engineer	80%	10 weeks	TBD
Integration Engineer (APIs/Adapters)	100%	10 weeks	TBD
Data Engineer (Streaming/ETL)	100%	10 weeks	TBD
Conversational AI Engineer	80%	10 weeks	TBD
Frontend/Mobile Integration Dev	60%	8 weeks	TBD
QA Automation Engineer	80%	10 weeks	TBD
DevSecOps Engineer	60%	10 weeks	TBD

5. Model Development, XAI, and MLOps (Phase 5: 4–6 weeks)

Phase 5 focuses on delivering production-ready predictive models and the MLOps pipelines that govern them. Using curated features from the feature store, we build churn prediction models leveraging behavioral signals, transactional trends, and engagement metrics. Propensity models estimate responses to offers, and next-best-action logic combines model outputs with eligibility rules, risk constraints, and business priorities. Model explainability is implemented via SHAP-like post-hoc explanations and interpretable techniques where feasible; we define explanation surfaces for business users and, when appropriate, customer-facing summaries. Bias testing evaluates protected and proxy attributes, with thresholds that trigger mitigation strategies such as reweighting or constrained optimization. The MLOps framework enforces traceability from data to model to decision, with automated retraining schedules, canary/shadow deployments, drift detection, rollback policies, and human-in-the-loop approvals for high-impact actions. These controls ensure transparency, fairness, and auditability in line with KFH policies and regulatory expectations.

Deliverable	Description	Format	Delivery Date
Churn, Propensity, NBA Models	Trained, validated models with XAI	Model Artifacts + Docs	Week 18–20
Bias & Fairness Report	Bias tests, thresholds, and mitigation plan	PDF	Week 19
MLOps Pipelines	CI/CD for models, registry, deployment patterns	Code + Runbooks	Week 18–20
Decisioning Ruleset	Business rules, eligibility, and thresholds	Config + Docs	Week 19–20

6. System Integration, Hardening, and HA/DR Validation (Phase 6: 3–4 weeks)

End-to-end integration establishes data flow continuity across channels, core systems, Customer 360, feature store, models, and orchestration. We execute comprehensive test suites that include functional, regression, integration, performance, and security testing. Performance tests validate throughput, latency, and autoscaling under expected and peak loads, with tuning at the application and platform layers to meet SLOs. Security hardening addresses IAM policies, encryption configurations, secrets rotation, vulnerability remediation, and pen test findings. HA/DR drills validate failover and failback for

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both active-active and active-passive scenarios, verifying RTO/RPO compliance and documenting runbooks. Observability is finalized with dashboards, alerts, and procedures for incident response and root cause analysis.

Milestone Name	Target Date	Success Criteria	Dependencies
E2E Integration Complete	Week 22–23	All interfaces validated; error budgets within limits	Phase 4 foundations
Performance & Scale Sign-off	Week 23	Latency/throughput SLOs met under peak load	Test data and load models
HA/DR Validation	Week 23–24	Documented RTO/RPO achieved in drills	Secondary site readiness
Security Hardening Sign-off	Week 24	No critical vulnerabilities; pen test remediated	Security review windows

7. UAT, Training, Go-Live, and Hypercare (Phase 7: 2–3 weeks)

User Acceptance Testing is conducted with business owners across personas using traceable test cases aligned to the problem-to-outcome matrix. Defects are triaged and resolved rapidly through a stabilized release branch. We deliver role-based training for marketing users, contact center agents, branch staff, and IT/Ops, covering operational runbooks, campaign setup, and model interpretation. Change management prepares internal communications, FAQs, and support workflows for a smooth rollout. The go-live plan sequences enablement by channel with controlled feature flags and rollback procedures. Hypercare provides elevated support, daily standups, and health dashboards to ensure stability and rapid response to incidents or change requests. Transition to steady-state operations includes handover to KFH teams, with knowledge transfer and final documentation baselines.

Deliverable	Description	Format	Delivery Date
UAT Plan & Test Cases	Persona-aligned test suites and acceptance criteria	Document	Week 24–25
Training & Enablement Pack	Role-based guides, videos, and hands-on labs	PDF/Video	Week 25
Go-Live Runbook	Cutover steps, monitoring, rollback, comms	Document	Week 25–26
Hypercare Report	Stabilization metrics and lessons learned	PDF	Week 27

8. Agile Cadence, Quality Assurance, and Change Control

The program runs on a standard two-week sprint cadence with planning, daily standups, mid-sprint checkpoints, and end-of-sprint demos and retrospectives. Stage-gate reviews conclude Phases 2, 3, 5, and 6 to confirm readiness for the next stage. Quality assurance spans unit, integration, contract, performance, and security testing, integrated into CI/CD pipelines with automated checks and manual exploratory testing where needed. The Definition of Done includes code and model reviews, documentation, observability hooks, security scans, and successful deployment to the target environment. Change control is governed by a joint Change Advisory Board; production-impacting changes require risk assessment, backout plans, and maintenance windows. Model changes follow a

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dedicated approval workflow under the model governance board, with documented rationale, XAI summaries, and bias assessments prior to promotion.

9. Communication Plan and Stakeholder Engagement

Clear, predictable communication underpins adoption and governance. Weekly status updates summarize progress, risks, decisions, and upcoming milestones. Executive steering meetings occur biweekly to address cross-functional dependencies and unblock decisions. Delivery working sessions bring together architects, engineers, and product owners to resolve technical issues in real time. A shared dashboard provides visibility into sprint burndown, defect metrics, model performance, and platform health. For stakeholder enablement, we schedule focused showcases of key capabilities such as the Robo Assistant, Customer 360 insights, and journey orchestration, ensuring that business leaders can validate outcomes and prioritize next increments.

10. Timeline at a Glance

The following plan outlines the phased delivery from discovery through go-live. Dates are expressed as relative weeks from project commencement (Week 1). Start and end markers may be adjusted collaboratively to align with KFH release windows and regulatory calendars.

Phase	Duration	Start Date	End Date	Key Deliverables
Phase 1: Kickoff & Governance	2 weeks	Week 1	Week 2	Governance framework, RACI, environments ready
Phase 2: Discovery & Personas	3–4 weeks	Week 3	Week 6	Persona journeys, outcomes matrix, source inventory, NFR baseline
Phase 3: Architecture & Integration Design	4–6 weeks	Week 7	Week 12	Target architecture, security design, integration specs, runbook outlines
Phase 4: Build Foundations & MVPs	8–10 weeks	Week 13	Week 22	OpenShift services, Customer 360 and feature store (initial), streaming pipelines, Robo Assistant MVP
Phase 5: Models & Orchestration	4–6 weeks	Week 18	Week 23	Churn/propensity/NBA models, XAI/bias governance, campaign journeys, MLOps pipelines
Phase 6: E2E Integration & Hardening	3–4 weeks	Week 22	Week 24	Performance tests, HA/DR validation, security hardening, observability
Phase 7: UAT, Training & Go-Live	2–3 weeks	Week 25	Week 27	UAT sign-off, training, production deployment, hypercare

Milestone Name	Target Date	Success Criteria	Dependencies

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Design Sign-off	Week 12	Architecture and security baseline approved	Completion of Phase 3
Integration Readiness	Week 18	APIs, adapters, and streaming endpoints operational in test	Phase 4 progress on integrations
Model Go/No-Go	Week 23	Performance, fairness, and XAI thresholds met	Model training and validation
Security Review Sign-off	Week 24	Pen test findings remediated; audit controls verified	Hardening and testing
UAT Sign-off	Week 26	Acceptance criteria met across personas	E2E integration completion
Production Go-Live	Week 27	Stable release deployed; monitoring green	UAT and CAB approvals

11. Risk Management and Mitigation

Risk management is continuous, with risks identified during planning and updated each sprint. We use a RAID log to track risks, assumptions, issues, and decisions, and we engage KFH stakeholders to ensure shared ownership of mitigations. Key risks include integration complexity across heterogeneous systems, data quality and lineage gaps, model bias or drift, performance under peak load, and changes in regulatory interpretation. We mitigate integration risk by using well-defined contracts, schema registries, and idempotent patterns, alongside early test harnesses and sandboxed stubs. Data risks are addressed through profiling, quality rules, lineage, and exception handling. Model risks are controlled via XAI, bias testing, controlled rollouts, and human-in-the-loop oversight. Performance risks are reduced through load testing, autoscaling, and capacity planning; compliance risk is mitigated through early and continuous involvement of Risk/Compliance and by embedding privacy-by-design controls.

Risk	**Likelihood (H/M/L)**	**Impact (H/M/L)**	**Mitigation Strategy**
Integration Complexity	M	H	Early contracts, API mocks, event schemas, incremental integration tests
Data Quality & Lineage Gaps	M	H	Data profiling, quality checks, lineage tracking, remediation workflows
Model Bias/Drift	M	H	Bias testing, drift monitoring, retraining schedules, governance approvals
Performance Degradation at Peak	M	H	Load testing, autoscaling, caching, capacity planning
Security & Compliance Findings	L	H	Secure SDLC, pen testing, IAM hardening, audit logging

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Change Management Resistance	M	M	Early engagement, training, showcases, change champions

12. Deliverables and Handover

Deliverables span architecture artifacts, working software, data pipelines, models, orchestration journeys, security baselines, runbooks, and training materials. Each deliverable is versioned in source control and linked to acceptance criteria. Handover includes operational documentation, monitoring dashboards, and knowledge transfer sessions for KFH teams. Post go-live, we conduct a lessons-learned review and finalize the sustainability plan for continuous improvement and model lifecycle operations.

Deliverable	Description	Format	Delivery Date
Target Architecture Package	Logical/physical diagrams, security and DR design	Diagrams + Docs	Week 12
Customer 360 & Feature Store	Data model, pipelines, golden profiles, feature catalog	Code + Docs	Week 18–22
Robo Assistant MVP	Conversational flows, RAG integration, hand-off	Code + KB + Docs	Week 18–22
Predictive Models & MLOps	Churn/propensity/NBA, XAI, registry, CI/CD	Artifacts + Runbooks	Week 20–23
Orchestration Journeys	Event-triggered campaigns, A/B tests, consent controls	Config + Playbooks	Week 20–23
HA/DR & Security Artifacts	DR runbooks, IAM policies, audit and logging	Docs + Config	Week 24
UAT & Training Materials	Test results, user guides, enablement content	Reports + PDFs	Week 25–26
Go-Live & Hypercare Report	Deployment checklist, stability metrics, handover	Runbook + Report	Week 27

This phased, agile approach ensures that KFH realizes incremental value at each stage while building a resilient foundation for long-term innovation. By anchoring delivery in OpenShift-based microservices, real-time streaming, a governed Customer 360 and feature store, explainable AI with bias controls, and enterprise-grade security and reliability, HYVE's engagement model provides a clear pathway from strategy to measurable outcomes and sustainable operations.

Solution Overview

The proposed solution delivers an integrated, secure, and scalable platform that unifies Kuwait Finance House's customer engagement across mobile, web, ATMs, kiosks, XTMs, and branch experiences. It is engineered around Red Hat OpenShift to provide container orchestration, service mesh security, autoscaling, and CI/CD discipline, while a real-time data backbone feeds a governed Customer 360 and feature store to power predictive analytics and next-best actions. The end result is a resilient capability stack that enables explainable, bias-governed AI to personalize experiences, prevent churn, and orchestrate omnichannel journeys with enterprise-grade reliability, observability, and compliance.

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1. Reference Architecture on Red Hat OpenShift

At the core of the solution is a microservices architecture deployed on Red Hat OpenShift. Each business capability—identity resolution, customer profile services, offer decisioning, conversational orchestration, campaign delivery, and channel adapters—is delivered as an independently deployable service. OpenShift provides declarative deployments, policy enforcement, and clusters spanning multiple availability zones to ensure high availability and rapid failover. A service mesh enables secure, observable communication between services via mutual TLS, fine-grained traffic management, retries, and circuit breakers. GitOps pipelines promote consistent, auditable releases from development to production, allowing frequent, low-risk updates to both application code and infrastructure-as-code. Containerization of AI services ensures that inference workloads scale elastically as traffic fluctuates, for instance during salary days or seasonal campaigns. Model-serving endpoints are deployed behind internal gateways with CPU and GPU autoscaling profiles, ensuring that p95 inference latencies remain within sub-150 millisecond budgets even under load. The platform supports blue/green, canary, and shadow deployment strategies, allowing new model versions and microservices to be validated safely before full promotion. Secrets management, config maps, and encrypted storage combine to minimize environment drift and protect sensitive credentials and keys.

Component	**Technology**	**Version**	**Purpose**
Container Orchestration	Red Hat OpenShift	Current LTS	Manage microservices, enforce policies, scale workloads
Service Mesh	OpenShift Service Mesh (Istio-based)	Current	mTLS, traffic shaping, retries, observability
API Management	Red Hat 3scale or equivalent	Current	Secure exposure, throttling, analytics, developer portal
CI/CD & GitOps	Tekton Pipelines + Argo CD	Current	Declarative deployments, promotion across environments
Event Streaming	Apache Kafka on OpenShift	Current	Real-time event backbone for channels and core systems
Lakehouse	Cloud/on-prem lakehouse (object + SQL)	Current	Customer 360 storage, batch analytics, governance
Feature Store	Enterprise/open-source feature store	Current	Curated, versioned features for online/offline use
Model Ops	MLflow/registry + inference gateways	Current	Model lifecycle, lineage, controlled rollouts
Observability	Prometheus, Grafana, EFK, Jaeger	Current	Metrics, logs, traces, alerting, SLO tracking
Security & IAM	SSO, RBAC/ABAC, secrets vault	Current	Identity, access controls, encryption, key rotation

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2. Integration Layer and Channel Connectivity

The integration layer provides a unified, secure interface to KFH's core banking systems, CRM, digital channels, and device endpoints. An API gateway mediates external and internal traffic with policies for authentication, authorization, throttling, and schema validation. Channel adapters encapsulate channel-specific nuances for mobile and web, while device connectors manage protocol and message formats for ATMs, kiosks, and XTMs. For legacy and batch-oriented systems, the solution employs an event-driven pattern: change data capture and event producers publish state transitions to Kafka topics, which are consumed by downstream services to update the Customer 360, trigger journeys, and refresh online features.

Event contracts and a schema registry enforce compatibility over time, enabling independent evolution of producers and consumers. To maintain resilience under transient failures, idempotent consumers and outbox patterns ensure exactly-once semantics at the business level. Integration to CRM unlocks interaction histories and campaign eligibility, while connectivity to the core provides up-to-date account and product entitlements. This integrated fabric ensures that a customer who begins an application on mobile can continue at an XTM or in a branch without repeating steps, and that service agents can view the same context as the AI assistant, improving first-contact resolution and consistency.

3. Data Platform: Customer 360 Lakehouse and Feature Store

The Customer 360 is the authoritative source of truth for identity, demographics, transactional history, behavioral signals, and service interactions. Golden profiles are created by resolving identities across banking channels and systems, employing deterministic and probabilistic matching rules with strict governance and lineage. The lakehouse architecture combines scalable object storage with SQL query engines, enabling both historical analytics and near-real-time enrichment. Data quality rules and lineage records are enforced at ingestion, and personally identifiable information is protected through masking, tokenization, and role- and attribute-based access controls.

A feature store bridges data to AI by curating features with consistent definitions, point-in-time correctness, and versioned transformations. Online feature serving provides low-latency access for real-time decisions, while the offline store supports batch training and performance analysis. Typical features include recent engagement frequency, transactional volatility, product eligibility flags, complaint history signals, and channel preference indicators. By centralizing feature computation and governance, the solution shortens time-to-model, improves reproducibility, and ensures that live decisions use the same semantics as training datasets, reducing drift and operational risk.

4. AI/ML Services and Decisioning

AI services focus on predicting churn risk, estimating propensity for offers, and determining next-best actions that respect eligibility, risk constraints, and customer preferences. Models are trained on curated features and validated against fairness, stability, and performance criteria prior to registration. Inference services expose REST or gRPC endpoints secured within the mesh, while a decisioning layer combines scores with business rules, consent and frequency caps, and channel availability to produce actionable recommendations. For example, a customer showing decreasing engagement and balance volatility may be prioritized for a retention outreach via in-app messaging followed by a relationship manager call, with thresholds and overrides approved through model governance workflows.

Explainability is embedded by design. For each decision, the system can produce an audit record enumerating key features and their contribution to the outcome, providing business stakeholders and auditors with transparent rationale. Where decisions affect eligibility or sensitive outcomes, human-in-the-loop approval is enforced. Models are monitored continuously for performance and drift, with retraining pipelines triggered by thresholds on accuracy decay, feature distribution shifts, or changes in regulatory policy. These controls ensure that the AI remains aligned with KFH's risk appetite and regulatory obligations while delivering measurable improvements in retention and conversion.

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5. Robo Assistant Experience and Channel Continuity

The AI-powered Robo Assistant orchestrates multilingual (Arabic and English) conversations across digital and assisted channels. It leverages retrieval-augmented generation to ground responses in curated, version-controlled knowledge bases covering product information, policies, and procedures. For transactional intents, the assistant invokes secure APIs to initiate and complete tasks such as card controls, bill payments, fee clarifications, or application status checks. Dialog state is persisted in a context service, allowing a customer to start a journey on mobile and continue through web, ATM guidance screens, or with a branch advisor without losing context.

Guardrails control what the assistant can answer and do, ensuring compliance and safety. Sensitive actions require strong authentication and step-up verification, while unsupported topics are deflected to a human agent with full context handover. In assisted channels, an agent assist mode presents suggested replies, next-best actions, and knowledge snippets tied to the live conversation, improving handling times and consistency. Latency targets are engineered for fluid, human-like interactions: end-to-end response times are designed to stay within two seconds for knowledge queries and under one second for cached or simple transactional confirmations.

6. Campaign Orchestration and Journey Management

A centralized journey orchestration capability coordinates event-triggered campaigns across push notifications, SMS, email, in-app banners, web personalization, ATM screens, and XTM prompts. Journeys are defined graphically with triggers based on real-time events, profile attributes, and model scores, and they enforce consent, channel preferences, and frequency caps at decision time. Experimentation is built into the orchestration flow, allowing A/B/n testing and audience holdouts to measure incremental lift accurately.

Marketers gain visibility into journey performance through role-based dashboards that present activation rates, conversion, average order value, and retention metrics by persona and channel. As journeys run, behavioral feedback loops update features and inform subsequent model training, enabling continuous optimization. This closed loop ensures that the most relevant message and next step are presented at the right time and channel, improving campaign ROI and customer satisfaction while minimizing contact fatigue.

7. Security, Privacy, and Compliance by Design

Security and privacy are foundational to the solution. Encryption in transit and at rest is mandatory for all data flows and storage layers. Strong identity and access management controls limit administrative and service access using least-privilege principles, and sensitive secrets are rotated regularly under automated policies. Comprehensive audit logging captures access events, data lineage, model decisions, and administrative changes, forming a regulator-ready trail. Data minimization, purpose limitation, and consent enforcement are implemented within orchestration and analytics workflows, ensuring that only approved data and features are used for specific decisions.

Compliance alignment includes data residency and localization where required, retention schedules that match regulatory mandates, and segregation of duties between development, operations, and analytics teams. The model governance board approves models for production based on documented performance, explainability, and bias assessment. When models influence customer outcomes, the system can generate human-readable explanations and provide mechanisms for review and override. These safeguards combine to protect customers, reduce operational risk, and foster trust while enabling rapid innovation.

8. Reliability, HA/DR, and Performance Engineering

Reliability is engineered across layers with multi-zone OpenShift clusters, stateless microservices, and fault-tolerant data paths. Stateful components are deployed in highly available configurations with synchronous replication where appropriate, and asynchronous replication across sites supports disaster

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recovery. The platform supports both active-active and active-passive patterns depending on workload characteristics and cost-performance tradeoffs. Failover drills validate recovery time and point objectives and ensure that runbooks and automation are effective under real-world conditions. Performance is addressed through capacity planning, autoscaling policies, and extensive performance testing. The streaming backbone is sized to handle peak ingest rates with headroom, and back-pressure management prevents cascading failures. Caching strategies reduce redundant calls to core systems for frequently accessed, non-sensitive data. For interactive experiences like the Robo Assistant and personalized UI components, response-time budgets are enforced end to end, with service-level objectives instrumented at each hop. This approach leads to predictable performance under load and efficient resource utilization.

Reliability Aspect	Target	Validation Method
Platform Uptime	99.9% or better	SLO dashboards, monthly reviews
DR RTO/RPO	RTO ≤ 60 minutes, RPO ≤ 5 minutes	Semi-annual DR drills, runbook sign-off
Chat Latency	p95 ≤ 2 seconds end-to-end	Synthetic probes, real-user monitoring
Streaming Freshness	< 60 seconds lag for key topics	Lag metrics, alert thresholds

9. Observability and FinOps

End-to-end observability enables rapid detection and resolution of issues across applications, data pipelines, and infrastructure. Metrics, logs, and distributed traces feed into dashboards with alerting based on service-level indicators. Golden signals such as latency, error rates, saturation, and traffic are tracked for critical services, and error budgets drive balanced decisions between feature velocity and reliability. For data and AI, specialized monitors track feature drift, model performance, and data quality rule violations, with automated tickets created when thresholds are breached.

FinOps practices are embedded to optimize cost without compromising performance. Workload right-sizing, autoscaling thresholds, and scheduled scale-downs reduce waste, while cost allocation tags provide transparency by team, environment, and capability. Periodic cost-performance reviews align resource usage with business value, surfacing opportunities to consolidate workloads, tune storage tiers, or leverage spot or burst capacity for non-critical training tasks. This disciplined approach ensures sustainable operations as adoption and data volumes grow.

10. Model and Data Governance with Explainability and Bias Controls

Governance spans data and models with an integrated approach. Data assets in the lakehouse carry ownership metadata, classification labels, retention policies, and approved use cases. Lineage graphs trace datasets from source to feature to model, allowing auditors and engineers to understand dependencies and impact. The feature catalog documents definitions, transformations, and quality metrics, enabling reuse and reducing duplication.

Model governance is enforced through a registry that tracks versions, training data snapshots, hyperparameters, evaluation metrics, and bias assessments. Promotion to production requires documented approvals, with risk-tiered review based on the model's business impact. Explainability reports summarize top drivers for predictions at both global and local levels, and customer-facing explanations are available for decisions that materially affect outcomes. Bias testing evaluates protected attributes and proxies, and mitigation strategies—such as constrained optimization, reweighting, or calibrated thresholds—are applied as needed. Ongoing monitoring watches for drift in data distributions and performance decay, automatically triggering retraining pipelines or rollback plans when thresholds are exceeded.

Governance Artifact	Owner	Review Cadence

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Data Dictionary & Lineage	Data Governance Lead	Quarterly or on schema change
Feature Catalog	Data & AI Lead	Monthly updates, sprint-based
Model Registry Entries	Model Governance Board	Pre-release and semi-annual audits
XAI & Bias Reports	Risk/Compliance Representative	Each major model version

In sum, the solution consolidates the capabilities KFH needs to deliver real-time, personalized, and trustworthy experiences across every channel. By standardizing on OpenShift for platform resilience and agility, implementing an event-driven integration fabric, operationalizing AI through a governed Customer 360 and feature store, and embedding rigorous security, compliance, and observability, KFH gains a future-ready foundation. This architecture not only addresses immediate priorities like churn prevention, cross-sell, and service automation, but also establishes a scalable operating model for continuous innovation in customer intelligence and journey orchestration.

Scope of Work

This section lists the details of the scope to be implemented including quantification of the deliverables, jobs , dashboards, etc. Should mention the activities and deliverables clearly

Scope	Activities
	•

Timeline

This is the solution timeline should include the various phases, how long each phase should take and expected activities in that phase.

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Scope of Work

This Scope of Work defines the full set of activities, deliverables, and acceptance criteria to implement Kuwait Finance House's AI-powered Robo Assistant and Customer 360 on Red Hat OpenShift. It translates KFH's business objectives into a structured program of work spanning persona-led design, omnichannel conversational experiences, unified customer data, predictive analytics, journey orchestration, real-time integrations, enterprise platform engineering, security and compliance, reliability, and enablement. Each workstream below includes measurable outputs, well-defined acceptance criteria, and governance checkpoints to ensure clarity, accountability, and on-time delivery.

1. Scope Overview and Objectives

The engagement will deliver a production-ready capability that unifies customer interactions across mobile, web, ATMs, kiosks, XTMs, and branches, powered by a governed Customer 360, operationalized AI for churn and next-best actions, and centralized journey orchestration. The platform will be engineered on Red Hat OpenShift using containerized microservices, service mesh security, CI/CD, and MLOps. Real-time streaming pipelines will keep profiles and features fresh, enabling truly contextual personalization and consistent service. Security, privacy, and explainability are embedded by design, with documented governance and audit artifacts supporting KFH's policies and regulatory obligations. Success is measured by increased self-service and conversion, reduced churn, faster time-to-resolution in assisted channels, and regulator-ready transparency for AI decisions.

2. Workstream A: Personas, Journeys, and KPIs

This workstream grounds delivery in the needs of priority personas, mapping problem statements to measurable outcomes and codifying them into journeys and acceptance criteria. The team will conduct interviews, analyze interaction data, and create target-state journey maps that enforce context continuity across channels. A problem-to-outcome matrix will anchor scope decisions and serve as the reference for UAT.

Persona	**Top Use Cases**	**Primary KPIs**
Retail Customer	Balance and transaction inquiries; card controls; bill payments; product discovery; onboarding	Self-service deflection rate ≥ 35%; task completion time ≤ 2 minutes for top 5 intents; CSAT ≥ 4.3/5
Premier/Wealth	Advisory prompts; portfolio-related queries; RM-assisted follow-up	Retention uplift ≥ 2–3 pts; RM productivity +15%; high-value client NPS +5
SME	Loan eligibility guidance; account opening; payment reconciliation	Onboarding cycle time −25%; application completion rate +20%; time-to-offer −15%
Branch Staff	Unified profile view; next-best-actions; service scripts	AHT −15%; cross-sell conversion +10%; first-contact resolution +8 pts
Contact Center	Contextual history; agent assist prompts; compliant responses	Deflection ≥ 30% of repetitive queries; FCR +10 pts; queue time −20%
Marketing	Event-triggered campaigns; A/B/n testing; attribution	Incremental lift +8–12%; CPA/CPE −15%; test cadence weekly

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IT/Ops	Predictable releases; observability; scalability	Change failure rate ≤ 5%; MTTR ≤ 30 minutes; availability ≥ 99.9%
Risk/Compliance	XAI; bias governance; audit trail	100% model decisions logged; bias thresholds adhered; audit readiness

Deliverables include persona journey maps, prioritized use cases, KPIs with baselines and targets, and a signed problem-to-outcome matrix. Acceptance criteria require senior stakeholder approval of journeys, traceability of user stories to measurable outcomes, and alignment with non-functional requirements (NFRs).

3. Workstream B: AI-Powered Robo Assistant (Omnichannel)

The Robo Assistant will provide secure, multilingual (Arabic/English) conversational support across channels, backed by retrieval-augmented generation grounded in approved knowledge assets and integrated with core systems for transactional intents. It will remember context across sessions and channels, and support seamless escalation to human agents with full context handover.

Capability	Scope Detail	Acceptance Criteria
Conversational Intents	40–60 high-frequency intents for Retail and SME; coverage for FAQs, transactions, status checks, and product discovery	≥ 90% intent recognition accuracy on validation set; ≥ 80% containment for top 10 intents
Languages	Arabic and English with locale-specific prompts and content	Language detection > 95%; parity of performance within ±5% across languages
Knowledge Grounding (RAG)	Responses grounded in version-controlled KB (policies, fees, product info)	Zero hallucination tolerance: 100% responses traceable to approved sources for regulated content
Transactions & Authentication	Secure API invocations for card controls, bill pay, application status; step-up auth for sensitive flows	p95 latency ≤ 2s for knowledge queries; ≤ 3s for transactional confirmations with auth
Context Continuity	Session/context service shared across channels; human handoff with full transcript	Handoff success ≥ 99%; context replay within 2s
Agent Assist	Suggested replies; next-best-actions; compliant scripts	AHT reduction ≥ 10%; agent adoption ≥ 75% within 60 days

Deliverables include conversational design assets, trained NLU/NLG models, RAG pipelines, channel SDKs/adapters, compliance guardrails, and handoff integrations. Test plans will cover intent accuracy, safety, latency, and multilingual parity.

4. Workstream C: Customer 360 and Feature Store

The Customer 360 will unify identities and interactions into golden profiles that drive personalization and analytics. The lakehouse will store historical and near-real-time data with quality, lineage, and privacy controls. A governed feature store will provide online and offline features for models and decisioning, with versioned transformations and point-in-time correctness.

Component	Scope Detail	Acceptance Criteria

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Identity Resolution	Deterministic and probabilistic matching across channels, CRM, and core banking	Precision/recall ≥ 0.97/0.95 on validation set; manual override workflows in place
Data Model	Customer, account, interaction, product, consent, and device entities	Logical and physical models approved; lineage captured for 100% curated datasets
Data Quality	Rules for completeness, validity, consistency; exception workflows	≥ 98% pass rate on critical rules; automated DQ alerts in place
Privacy & Access	RBAC/ABAC; masking/tokenization for PII; consent enforcement	No high/critical violations in privacy review; access logs enabled
Feature Store	100–150 curated features (engagement, eligibility, risk, behavior)	Feature catalog published; online latency ≤ 50ms p95; offline reproducibility verified

Deliverables include data models, pipeline code, golden profile jobs, data quality rulebooks, feature engineering pipelines, and a feature catalog with ownership and documentation.

5. Workstream D: Predictive Analytics and Decisioning

Predictive models will address churn risk, propensity for offers, and next-best actions. A decisioning layer will combine scores with rules, constraints, and consent to drive contextual recommendations in real time. Explainable AI (XAI) and bias governance will ensure transparency and fairness, with audit records for each decision.

Model/Asset	**Scope Detail**	**Acceptance Criteria**
Churn Model	Supervised learning with behavior, transaction, and service features	AUC ≥ 0.78 in validation; calibration error ≤ 3%; XAI reports delivered
Propensity Models	Offer-level response likelihood for key products	Top-decile lift ≥ 3x baseline; stability across personas within ±10%
Next-Best-Action	Policy layer combining scores, eligibility, risk, and caps	Business rules externalized; decision latency ≤ 150ms p95
Re-Engagement Playbooks	Dormancy triggers, KYC reminders, educational nudges	Treatment uplift ≥ 8–12% in A/B tests; consent and frequency caps enforced
XAI & Bias Testing	Global and local explanations; protected attribute analysis	All models pass bias thresholds; governance board approvals logged

Deliverables include model artifacts, training and evaluation reports, XAI summaries, bias/fairness assessments, decision policies, shadow/canary deployment configs, and monitoring dashboards for performance and drift.

6. Workstream E: Omnichannel Campaign Orchestration

A centralized journey orchestration capability will trigger campaigns based on events, profile attributes, and model scores, enforcing consent and channel preferences. It will support A/B/n testing, holdouts, and attribution reporting, with dashboards for marketers and business owners.

Capability	**Scope Detail**	**Acceptance Criteria**

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Journey Builder	Visual definition of multi-step, cross-channel journeys	10–15 production journeys configured; templated best practices
Triggers & Audiences	Event- and attribute-based triggering with model scores	End-to-end lag from trigger to action ≤ 60s; schema contracts validated
Experimentation	A/B/n with holdouts, guardrails, and auto-allocation	Statistically valid tests weekly; attribution model documented
Dashboards	Role-based views for activation, conversion, revenue	Near-real-time updates; data dictionary and SLAs published

Deliverables include configured journeys, experimentation framework, consent and preference enforcement, and role-based dashboards with documentation for key metrics and attribution logic.

7. Workstream F: Integrations and Data Pipelines

The solution integrates with core banking, CRM, digital channels, ATMs, kiosks, XTMs, and the data lake through an API gateway and real-time streaming. Adapters will encapsulate channel and device-specific protocols, while change data capture and event producers maintain freshness.

Interface	**Integration Pattern**	**Purpose**
Mobile/Web Channels	REST APIs + SDKs	Conversational UI, personalization, and journey triggers
CRM	REST/GraphQL + batch sync	Interaction history, campaign eligibility, and consent
Core Banking	Secure APIs + outbox/C DC events	Entitlements, balances, transactions, and status updates
ATMs/Kiosks/XTMs	Device connectors + message brokers	Context prompts, journey continuation, on-screen guidance
Data Lake	Streaming (Kafka) + batch ingestion	Historical analytics, 360 enrichment, feature backfills

Deliverables include API specs, schema registry entries, adapter implementations, streaming topics with governance, and test harnesses. Acceptance requires contract tests, idempotent patterns, back-pressure handling, and observability for retries and DLQs.

8. Workstream G: OpenShift Platform, CI/CD, and MLOps

The platform will be delivered on Red Hat OpenShift with service mesh, GitOps-driven CI/CD, and observability. MLOps will standardize the model lifecycle from training to deployment with governance and auditability.

Component	**Technology**	**Purpose**
Orchestration	Red Hat OpenShift	Containerized microservices and policy enforcement
Service Mesh	OpenShift Service Mesh (Istio-based)	mTLS, traffic management, and observability
API Management	3scale or equivalent	Secure exposure, throttling, analytics
CI/CD & GitOps	Tekton + Argo CD	Declarative pipelines, environment promotions
Streaming	Apache Kafka	Real-time event backbone

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MLOps	MLflow/registry + inference gateways	Model registry, deployment, lineage
Observability	Prometheus, Grafana, EFK, Jaeger	Metrics, logs, traces, SLO dashboards

Deliverables include cluster namespaces, policies, pipelines, deployment manifests, secrets management, observability dashboards, and MLOps workflows for training, validation, canary/shadow rollout, and drift-triggered retraining.

9. Workstream H: Security, Privacy, and Compliance

Security and privacy controls will be embedded in the platform and data flows. The scope includes encryption, IAM with RBAC/ABAC, network segmentation, secrets management, audit logging, data minimization, and consent enforcement. Governance will produce the artifacts needed for internal and external review.

Artifact	Description	Acceptance Criteria
Security Architecture	Controls for encryption, IAM, segmentation, secrets	Approved by Security & Compliance; no critical gaps
IAM Policies	Role and attribute-based access with least privilege	Access reviews complete; SoD maintained; audit logs enabled
Audit & Logging	End-to-end logs for access, data lineage, and decisions	100% key events logged; retention and immutability configured
Privacy Controls	Data minimization, masking/tokenization, consent	Privacy review passed; policy mappings documented
Model Governance	Registry, approvals, XAI, bias testing, monitoring	Governance board approvals; bias thresholds met

Deliverables include security design, IAM configurations, audit and logging setup, privacy mappings, model governance playbooks, and evidence packs aligned to KFH policy.

10. Workstream I: Reliability (HA/DR) and Performance Engineering

Reliability and performance will be validated through structured testing and documentation. The solution will support multi-zone availability, defined RTO/RPO, and workload-specific failover strategies. Performance SLOs will be established and tested for the assistant, streaming, decisioning, and APIs.

Validation Area	Target	Test/Validation Method
Availability	≥ 99.9% platform uptime	Chaos tests; SLO dashboards; monthly reviews
DR Objectives	RTO ≤ 60 min; RPO ≤ 5 min	Semi-annual drills; runbook execution timing
Chat Latency	p95 ≤ 2s knowledge; ≤ 3s transactional	Synthetic probes; real-user monitoring
Streaming Lag	≤ 60s for key topics	Kafka lag metrics; alert thresholds
Scale Behavior	Stable under 2x peak load	Load tests; autoscaling and capacity plans

Deliverables include HA/DR design, failover runbooks, performance test plans, tuning recommendations, and final sign-off reports.

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11. Workstream J: Documentation, Training, and Knowledge Transfer

Comprehensive documentation and enablement ensure sustainability post go-live. Materials will cover architecture, operations, runbooks, campaign orchestration, model interpretation, and security posture. Training will be role-specific and include hands-on labs where applicable.

Deliverable	Audience	Format
Solution Design Package	Architects, IT/Ops	Diagrams + documents
Operations Runbooks	IT/Ops, SRE	Step-by-step guides
Campaign & Journey Guide	Marketing	Playbooks + videos
Model & XAI Handbook	Data Science, Risk/Compliance	Reports + demos
Platform & CI/CD Guide	Dev/DevOps	Repos + tutorials
User Training & Labs	Agents, Branch Staff, Marketers	Instructor-led + eLearning

Acceptance criteria include completed training sessions with attendance tracking, knowledge checks, repository-based documentation, and recorded enablement sessions for future reference.

12. Out-of-Scope and Assumptions

The scope excludes unrelated legacy system upgrades, non-banking external data provider contracts, and large-scale replatforming of core banking beyond required integration adapters. Model coverage is focused on churn, propensity, and next-best actions; credit risk underwriting models are not included. Channel UI redesign outside assistant integration and personalization widgets is out of scope. It is assumed that KFH will provide timely access to systems and subject matter experts, that required environments and network connectivity will be provisioned as planned, and that third-party licenses (if any) are procured by KFH unless explicitly stated otherwise.

13. Acceptance Criteria and Success Metrics

Each workstream has explicit acceptance criteria aligned to business outcomes and non-functional targets. The following table summarizes cross-workstream criteria that govern overall acceptance.

Area	Metric/Threshold	Acceptance Evidence
Self-Service & Deflection	≥ 30–35% for top intents	UAT and pilot analytics reports
Churn Reduction	≥ 2–4 pt improvement in target segments	A/B test and cohort analysis
Offer Conversion	≥ 8–12% incremental lift over baseline	Experimentation dashboard exports
Availability	≥ 99.9% over pilot period	SLO/SLI dashboard and incident logs
Security & Privacy	Zero critical findings; audit-ready evidence	Pen test results; policy mappings
Explainability & Bias	All production models pass thresholds	XAI and bias reports; approvals
Performance	Chat p95 ≤ 2–3s; decision ≤ 150ms p95	Load test and RUM dashboards

14. Deliverables Summary

The table below consolidates major deliverables by category with indicative formats. Each item will be versioned in source control and linked to acceptance criteria in the backlog.

Category	Deliverable	Format
Personas & Journeys	Journey maps; problem-to-outcome matrix; KPIs	PDFs, spreadsheets
Robo Assistant	Conversational flows; models; RAG pipelines; guardrails	Code, configs, KB

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Customer 360	Data models; pipelines; golden profiles; DQ rules	Code, docs
Feature Store	Feature catalog; transformations; online/offline stores	Code, catalog
AI & Decisioning	Churn/propensity/NBA models; XAI/bias reports; policies	Artifacts, reports
Orchestration	Event-triggered journeys; A/B/n framework; dashboards	Config, dashboards
Integrations	API specs; adapters; schema registry; streaming topics	Specs, code
Platform & DevOps	OpenShift configs; CI/CD; observability	Manifests, pipelines
Security & Compliance	IAM policies; audit/logging; privacy mappings; governance playbooks	Docs, configs
Reliability & Performance	HA/DR design; runbooks; load test results	Docs, reports
Enablement	Training materials; user guides; recorded sessions	PDFs, videos

This Scope of Work ensures a cohesive, end-to-end delivery that addresses KFH's immediate priorities while establishing a durable foundation for future innovation. By anchoring execution in persona-driven outcomes, governed data and AI, robust integrations, OpenShift-based platform engineering, and rigorous security and reliability, the program provides a clear path to measurable business impact and sustainable operations.

Timeline

The delivery timeline is structured around outcome-driven, two-week sprints with stage-gate checkpoints to ensure traceable progress, timely risk mitigation, and business-readiness at every phase. Our plan spans discovery through production go-live and hypercare, includes controlled overlaps where parallelization reduces elapsed time, and allocates explicit windows for integration, security reviews, and model governance approvals. Dates are expressed as relative weeks from project commencement (Week 1) and will be refined collaboratively to align with KFH's release calendar, change windows, and regulatory review schedules.

1. Timeline Overview and Planning Assumptions

The program follows a sequenced yet partially overlapping set of phases that accelerate value while preserving quality and compliance. Core assumptions include two-week sprints, shared KFH/HYVE governance, timely access to environments and systems, and availability of business stakeholders for reviews. We embed continuous testing and documentation from the outset to avoid downstream bottlenecks. Where feasible, activities such as model development and campaign configuration begin before all integrations are finalized by using synthetic data, stubs, and sandboxes—then converge on production-grade interfaces in later phases.

Phase	Duration	Start Date	End Date	Key Deliverables
Phase 1: Kickoff, Governance & Planning	2 weeks	Week 1	Week 2	Governance framework, RACI, communication plan,

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				environments/access readiness, initial backlog
Phase 2: Discovery & Persona Outcomes	3–4 weeks	Week 3	Week 6	Persona journey maps, problem-to-outcome matrix, source system inventory, NFR & compliance baseline
Phase 3: Architecture & Integration Design	4–6 weeks	Week 7	Week 12	Target architecture, API/streaming designs, security baseline, HA/DR strategy, runbook outlines
Phase 4: Build Foundations & MVPs	8–10 weeks	Week 13	Week 22	OpenShift platform services, Customer 360 (initial), feature store (initial), streaming pipelines (initial), Robo Assistant MVP, orchestration baseline
Phase 5: Models, XAI & Campaign Orchestration	4–6 weeks (overlaps)	Week 18	Week 23	Churn/propensity/NBA models, XAI/bias reports, MLOps pipelines, event-triggered journeys, experimentation framework
Phase 6: E2E Integration, Performance & Security Hardening	3–4 weeks (overlaps)	Week 22	Week 24	E2E integration validation, performance testing, HA/DR drills, security remediation and sign-off, observability finalization
Phase 7: UAT, Training, Go-Live & Hypercare	2–3 weeks	Week 25	Week 27	UAT sign-off, role-based training, production cutover, hypercare, handover

Planning assumptions include: alignment to KFH change freezes and peak periods (e.g., salary days), model governance board availability for review cycles, and security review windows for penetration testing and remediation. Where procurement or third-party licensing is required, lead times are factored into pre-sprint readiness activities and tracked in the RAID log.

2. Phase Detail and Key Activities

2.1 Phase 1: Kickoff, Governance & Planning (Weeks 1–2)

Phase 1 mobilizes the program and sets the cadence for delivery and decision-making. Activities include forming the steering committee, delivery working group, and model governance board; confirming

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scope, success metrics, and communication plans; and establishing environments and access paths for development and testing. The backlog is seeded with epics and user stories mapped to persona outcomes, and quality gates are defined for code, data, security, and model artifacts. By the end of Week 2, the governance rhythm is operational, with baselined artifacts stored in the collaboration workspace and repos initialized for infrastructure-as-code, microservices, and data pipelines.

2.2 Phase 2: Discovery & Persona Outcomes (Weeks 3–6)

In Phase 2, the team conducts structured workshops and interviews with Retail, Premier/Wealth, SME, Branch, Contact Center, Marketing, IT/Ops, and Risk/Compliance stakeholders to validate top tasks and measurable outcomes. Current and future-state journeys are captured with emphasis on context continuity across mobile, web, ATMs, kiosks, XTMs, and branches. Parallel data discovery compiles a source system inventory, event catalogs, and integration SLAs. Non-functional requirements (NFRs) covering performance, HA/DR, privacy, and data localization are baselined. Deliverables include a signed problem-to-outcome matrix and prioritized use cases driving the initial MVP scope for the Robo Assistant, Customer 360, and orchestration.

2.3 Phase 3: Architecture & Integration Design (Weeks 7–12)

Phase 3 delivers the target architecture centered on OpenShift microservices, service mesh, API gateway, and Kafka-based streaming. Security design encompasses IAM, RBAC/ABAC, secrets management, encryption, and audit logging. HA/DR strategies are selected per workload based on RTO/RPO targets. Data architecture defines the Customer 360 lakehouse and governed feature store with lineage and data quality checkpoints. Integration specifications are produced for mobile/web, CRM, core banking, ATMs, kiosks, XTMs, and the data lake. Stage-gate sign-off at Week 12 confirms readiness to build with clear interfaces, contracts, and security controls.

2.4 Phase 4: Build Foundations & MVPs (Weeks 13–22)

Foundational platform capabilities and MVP features are implemented through multiple two-week sprints. OpenShift clusters are configured with namespaces, policies, ingress/egress rules, and service mesh. CI/CD pipelines support automated builds, scans, and progressive delivery. Integration adapters and stubs connect channels, CRM, and core systems; streaming pipelines begin capturing real-time signals. The initial Customer 360 data model is finalized, golden profiles are created for target cohorts, and the feature store exposes curated features for models and decisioning. The Robo Assistant MVP covers high-frequency intents in Arabic and English with retrieval-augmented responses grounded in approved knowledge; secure session handling and human handoff are implemented. Orchestration enables event-triggered journeys with consent and frequency caps. Regular demos validate progress with business stakeholders, and performance budgets for conversational interactions are monitored in test environments.

2.5 Phase 5: Models, XAI & Campaign Orchestration (Weeks 18–23, overlaps Phase 4)

Model development starts as soon as curated features are available, initially using historical and synthetic datasets, then converging on live event streams. Churn, propensity, and next-best-action models are trained, validated, and registered with accompanying XAI and bias assessments. MLOps pipelines automate training, evaluation, deployment, canary/shadow testing, and drift monitoring. Campaign orchestration is configured for priority journeys with A/B/n testing and holdouts. Alignment with Risk/Compliance ensures that explanation policies and bias thresholds meet governance standards. A go/no-go checkpoint for model promotion and journey activation is scheduled in Week 23.

2.6 Phase 6: E2E Integration, Performance & Security Hardening (Weeks 22–24, overlaps Phase 5)

The focus shifts to end-to-end validation across channels, integrations, Customer 360, feature store, models, and orchestration. Performance testing validates throughput, latency, and autoscaling,

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including peak-load simulations and failover scenarios. Security hardening addresses identified vulnerabilities, IAM policy refinements, secrets rotation, and pen test remediation. HA/DR drills verify RTO/RPO targets and document runbooks for failover and failback. Observability dashboards and alerts are finalized, and readiness reviews confirm stability and compliance for UAT commencement.

2.7 Phase 7: UAT, Training, Go-Live & Hypercare (Weeks 25–27)

User Acceptance Testing executes persona-aligned test cases tied to the problem-to-outcome matrix. Defects are triaged and resolved via a stabilized release branch. Training and enablement provide role-based guidance for marketers, agents, branch staff, and IT/Ops. The cutover plan sequences go-live by channel, using feature flags and rollback procedures to de-risk deployment. Hypercare delivers elevated monitoring and daily standups to ensure quick issue resolution and knowledge transfer. Final handover includes operational documentation, dashboards, and a lessons-learned report to inform continuous improvement.

3. Milestones, Stage Gates, and Readiness Criteria

Milestones anchor decision points and control transitions between phases. Each milestone requires documented evidence, including test results, approvals, and governance artifacts. The stage gates ensure alignment between business outcomes, technical readiness, and compliance.

Milestone Name	Target Date	Success Criteria	Dependencies
Design Sign-off	Week 12	Target architecture, security baseline, integration specs approved	Completion of Phase 3 artifacts
Integration Readiness	Week 18	APIs/adapters and key streaming topics operational in test; contract tests passed	Phase 4 integration progress
Model Go/No-Go	Week 23	Performance, fairness, and XAI thresholds met; governance approvals logged	Model training/validation; feature store stability
Security Review Sign-off	Week 24	Pen test findings remediated; IAM, logging, and encryption verified	Security hardening and testing
UAT Sign-off	Week 26	Acceptance criteria met across personas; defects within agreed thresholds	E2E integration completion
Production Go-Live	Week 27	Stable release deployed; monitoring green; CAB approvals complete	UAT sign-off, change window availability

Readiness criteria for each stage are tracked in the backlog and linked to test evidence, documentation, and sign-off records. No milestone is closed without explicit approval from designated KFH stakeholders and the HYVE delivery lead.

4. Dependencies and Critical Path

The critical path runs through environment readiness (Phase 1), architecture sign-off (Phase 3), foundation builds and data readiness (Phase 4), model validation (Phase 5), and hardening/UAT (Phase

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6–7). While model and campaign work can begin before all integrations are live, production promotion requires completed E2E validation and governance approvals. Key dependencies include access to core APIs and CRM data, data lake connectivity, security review windows, and channel development slots for embedding the Robo Assistant and personalization components.

Dependency	Owner	Needed By	Impact if Delayed
OpenShift Cluster Namespaces & Access	IT/Ops	Week 2	Delays pipeline setup, slows Phase 4 starts
API Contracts for Core & CRM	Integration Teams	Week 8	Slips adapter builds; increases rework risk
Kafka Infrastructure & Schema Registry	Platform/Data	Week 12	Blocks streaming pipelines and near-real-time features
Knowledge Base Content (Approved)	Business/Compliance	Week 16	Limits RAG coverage; reduces assistant containment
Penetration Testing Window	Security	Week 23	Pushes security sign-off; risks go-live date
CAB/Change Window	IT Governance	Week 27	Moves production deployment outside planned window

We maintain a living RAID log that captures status, mitigation, and decision ownership for each dependency. The steering committee reviews critical-path items weekly and expedites decisions when needed.

5. Resource Loading and Collaboration Cadence

Resourcing aligns to sprint cadence, with cross-functional teams spanning platform engineering, integrations, data engineering, data science, security, QA, and change management. Collaboration includes daily standups, weekly status updates, biweekly steering meetings, and sprint reviews with business stakeholders. To minimize context switching, dedicated roles are assigned for model governance and security enablement during Phases 5 and 6.

Resource	Allocation %	Duration	Cost
OpenShift Platform Engineer	60–80%	Weeks 1–22	TBD
Integration Engineer (APIs/Adapters)	80–100%	Weeks 7–24	TBD
Data Engineer (Streaming/ETL)	80–100%	Weeks 7–24	TBD
Data Scientist/ML Engineer	60–100%	Weeks 13–23	TBD
Conversational AI Engineer	60–80%	Weeks 13–22	TBD
QA/Automation Engineer	60–100%	Weeks 13–27	TBD
DevSecOps Engineer	50–70%	Weeks 7–24	TBD
Security & Compliance Lead	30–50%	Weeks 7–27	TBD
Training & Change Manager	30–60%	Weeks 20–27	TBD

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Resourcing flexes in response to milestone risk: for example, additional QA and platform capacity are reserved during performance testing and hardening to accelerate remediation. Knowledge transfer is embedded throughout to reduce reliance on specialized roles post go-live.

6. Quality Gates, Testing Strategy, and Evidence

Quality gates apply at multiple levels. Unit, contract, and integration tests are executed in CI/CD; performance tests validate p95 and p99 latencies for chat flows and decisioning; security scanning and pen testing verify compliance; and data quality rules enforce completeness and validity for Customer 360 and feature store pipelines. For models, MLOps pipelines enforce reproducibility, versioning, and bias testing. Evidence for each gate is captured in dashboards and reports, referenced in milestone checklists, and retained for audit.

Validation Area	Metric/Threshold	Evidence
Conversational Latency	p95 ≤ 2s (knowledge), ≤ 3s (transactional)	Load test reports, RUM dashboards
Decisioning Latency	p95 ≤ 150ms	Synthetic probes, tracing
Streaming Freshness	Kafka lag ≤ 60s for key topics	Lag metrics, alerts
Data Quality	≥ 98% pass rate on critical rules	DQ dashboards, exception logs
Model Performance	Churn AUC ≥ 0.78; calibrated	Eval reports, registry entries
Security	Zero critical vulns; mTLS enforced	Scan reports, mesh policies
HA/DR	RTO ≤ 60m, RPO ≤ 5m	DR drill results, runbooks

7. Schedule Risks and Contingencies

We proactively manage schedule risk with buffers at milestone boundaries, parallel workstreams where safe, and early test harnesses to de-risk late integrations. Key risks include integration complexity across heterogeneous systems, regulatory review delays, data quality remediation taking longer than expected, and performance tuning under peak load. Contingencies include scope shaping for non-critical intents in MVP, phased channel rollout with feature flags, and pre-approved change windows.

Risk	Likelihood (H/M/L)	Impact (H/M/L)	Mitigation Strategy
Integration Delays	M	H	Early contracts, mocks, schema registry, incremental E2E tests
Data Quality Gaps	M	H	DQ rules early, remediation sprints, lineage and exception flows
Regulatory Sign-off Timing	M	M	Reserve review windows, provide early evidence packs, pre-read sessions
Model Fairness Thresholds Not Met	L	M	Alternate feature sets, reweighting, constrained optimization, human-in-loop
Performance Under Peak Load	M	H	Load tests, autoscaling, caching, capacity planning and tuning

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Change Freeze/Release Windows	L	M	Flexible deployment plan, feature flags, blue/green fallback

If a critical-path risk materializes, the steering committee will invoke a fast-track decision process to adjust scope, reassign capacity, or shift milestones. Non-critical features may be scheduled for a post-go-live incremental release without jeopardizing core outcomes.

8. Alternative Scenarios and Optimizations

To accommodate evolving priorities or external constraints, we outline optional scenario adjustments. An accelerated scenario compresses Phase 4 and Phase 5 by increasing parallel squads and pre-validating integrations with enhanced stubs, targeting earlier MVP demonstration in Week 18. A risk-reduced scenario extends Phase 6 by one week to incorporate a second performance and security iteration, providing additional assurance ahead of go-live. In both cases, the core milestone sequence remains intact, and any scope trade-offs are documented against the problem-to-outcome matrix to protect business value.

9. Transition to Steady State

Post hypercare, we transition into steady-state operations with agreed SLAs, monitoring, and model lifecycle processes. A quarterly optimization roadmap captures backlog items not critical for initial go-live—such as additional intents, extended feature engineering, or new journey variants—and ties them to measurable KPIs. This ensures sustained momentum and continuous improvement without disrupting operational stability.

In summary, the proposed 27-week timeline balances speed and rigor through phased delivery, selective overlaps, and disciplined stage gates. By aligning activities to persona outcomes, embedding security and compliance from the start, and enforcing quality gates with clear evidence, KFH gains a predictable path to production—culminating in a secure, explainable, and high-performing platform for the AI-powered Robo Assistant and Customer 360 capabilities on Red Hat OpenShift.

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